FREELANCE MARKETPLACE

Database Schema

MySQL 8.0 Database Design

Technology Stack: Python + Django + MySQL

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Academic Project Documentation

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1. Database Overview

This document provides the complete database schema for the Freelance Marketplace application. The database is designed for MySQL 8.0 and follows Django ORM conventions.

1.1 Database Configuration

Property	Value
Database Engine	MySQL 8.0
Character Set	utf8mb4
Collation	utf8mb4_unicode_ci
Storage Engine	InnoDB
Primary Key Type	UUID (CHAR(36))
Timestamp Fields	created_at, updated_at (auto-managed)

1.2 Table Summary

Table Name	Description	Row Estimate
users	User accounts and authentication	10,000+
freelancer_profiles	Freelancer extended profiles	5,000+
client_profiles	Client extended profiles	5,000+
categories	Project categories	50+
skills	Skill tags	500+
projects	Project listings	50,000+
project_attachments	Project file attachments	100,000+
proposals	Freelancer proposals/bids	200,000+
contracts	Active contracts	20,000+
milestones	Contract milestones	60,000+
deliverables	Work deliverables	100,000+
conversations	Chat conversations	30,000+
messages	Chat messages	500,000+
transactions	Payment transactions	100,000+
escrow_accounts	Escrow holdings	20,000+
freelancer_wallets	Freelancer wallets	5,000+
reviews	User reviews	40,000+
notifications	User notifications	1,000,000+

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2. Table Schemas

2.1 Users Table

Table: users

Column	Data Type	Null	Key	Description
id	CHAR(36)	NO	PK	UUID primary key
email	VARCHAR(254)	NO	UQ, IDX	User email (unique)
username	VARCHAR(150)	NO	UQ	Username
password	VARCHAR(128)	NO		Hashed password
first_name	VARCHAR(150)	YES		First name
last_name	VARCHAR(150)	YES		Last name
role	ENUM	NO	IDX	freelancer/client/admin
avatar	VARCHAR(255)	YES		Avatar file path
phone	VARCHAR(20)	YES		Phone number
is_active	BOOLEAN	NO		Account active status
is_verified	BOOLEAN	NO		Email verified
is_staff	BOOLEAN	NO		Staff access
date_joined	DATETIME	NO		Registration date
last_login	DATETIME	YES		Last login timestamp
created_at	DATETIME	NO		Record creation
updated_at	DATETIME	NO		Last update

2.2 Freelancer Profiles Table

Table: freelancer_profiles

Column	Data Type	Null	Key	Description
id	CHAR(36)	NO	PK	UUID primary key
user_id	CHAR(36)	NO	FK, UQ	Reference to users.id
title	VARCHAR(100)	NO		Professional title
bio	TEXT	NO		Biography (max 2000)
hourly_rate	DECIMAL(10,2)	NO		Hourly rate in USD
skills	JSON	YES		Array of skill names
experience_years	INT UNSIGNED	NO		Years of experience
portfolio_url	VARCHAR(200)	YES		Portfolio website URL
availability	VARCHAR(50)	NO		Current availability
total_earnings	DECIMAL(12,2)	NO		Total earnings (default 0)

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avg_rating	DECIMAL(3,2)	NO		Average rating (0-5)
total_reviews	INT UNSIGNED	NO		Number of reviews
completed_projects	INT UNSIGNED	NO		Completed project count
created_at	DATETIME	NO		Record creation
updated_at	DATETIME	NO		Last update

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2.3 Projects Table

Table: projects

Column	Data Type	Null	Key	Description
id	CHAR(36)	NO	PK	UUID primary key
client_id	CHAR(36)	NO	FK, IDX	Reference to users.id
category_id	CHAR(36)	YES	FK, IDX	Reference to categories.id
title	VARCHAR(200)	NO	IDX	Project title
slug	VARCHAR(250)	NO	UQ	URL-friendly slug
description	TEXT	NO	FT	Detailed description
skills_required	JSON	YES		Array of required skills
budget_type	ENUM	NO		fixed/hourly/milestone
budget_min	DECIMAL(10,2)	NO		Minimum budget
budget_max	DECIMAL(10,2)	NO		Maximum budget
experience_level	ENUM	NO		entry/intermediate/expert
estimated_duration	VARCHAR(50)	NO		Duration estimate
status	ENUM	NO	IDX	Project status
deadline	DATE	YES		Project deadline
published_at	DATETIME	YES		When published
views_count	INT UNSIGNED	NO		View counter
proposals_count	INT UNSIGNED	NO		Proposal counter
is_featured	BOOLEAN	NO		Featured project flag
is_urgent	BOOLEAN	NO		Urgent project flag
created_at	DATETIME	NO	IDX	Record creation
updated_at	DATETIME	NO		Last update

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2.4 Proposals Table

Table: proposals

Column	Data Type	Null	Key	Description
id	CHAR(36)	NO	PK	UUID primary key
project_id	CHAR(36)	NO	FK, IDX	Reference to projects.id
freelancer_id	CHAR(36)	NO	FK, IDX	Reference to users.id
cover_letter	TEXT	NO		Proposal cover letter
bid_amount	DECIMAL(10,2)	NO		Bid amount in USD
estimated_duration	VARCHAR(50)	NO		Estimated timeline
status	ENUM	NO	IDX	pending/accepted/rejected
is_viewed	BOOLEAN	NO		Viewed by client flag
viewed_at	DATETIME	YES		When viewed
created_at	DATETIME	NO	IDX	Record creation
updated_at	DATETIME	NO		Last update

Unique Constraint: UNIQUE(project_id, freelancer_id)

2.5 Contracts Table

Table: contracts

Column	Data Type	Null	Key	Description
id	CHAR(36)	NO	PK	UUID primary key
project_id	CHAR(36)	NO	FK, UQ	Reference to projects.id
proposal_id	CHAR(36)	NO	FK, UQ	Reference to proposals.id
client_id	CHAR(36)	NO	FK, IDX	Reference to users.id (client)
freelancer_id	CHAR(36)	NO	FK, IDX	Reference to users.id (freelancer)
title	VARCHAR(200)	NO		Contract title
description	TEXT	NO		Contract description
total_amount	DECIMAL(10,2)	NO		Total contract value
status	ENUM	NO	IDX	active/completed/cancelled
start_date	DATE	NO		Contract start date
end_date	DATE	YES		Contract end date
terms_accepted	BOOLEAN	NO		Terms acceptance flag
created_at	DATETIME	NO		Record creation
updated_at	DATETIME	NO		Last update

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2.6 Milestones Table

Table: milestones

Column	Data Type	Null	Key	Description
id	CHAR(36)	NO	PK	UUID primary key
contract_id	CHAR(36)	NO	FK, IDX	Reference to contracts.id
title	VARCHAR(200)	NO		Milestone title
description	TEXT	NO		Milestone description
amount	DECIMAL(10,2)	NO		Milestone amount
due_date	DATE	NO		Due date
order	INT UNSIGNED	NO		Sort order
status	ENUM	NO	IDX	pending/in_progress/submitted/approved/paid
submitted_at	DATETIME	YES		Submission timestamp
approved_at	DATETIME	YES		Approval timestamp
created_at	DATETIME	NO		Record creation
updated_at	DATETIME	NO		Last update

2.7 Transactions Table

Table: transactions

Column	Data Type	Null	Key	Description
id	CHAR(36)	NO	PK	UUID primary key
user_id	CHAR(36)	NO	FK, IDX	Reference to users.id
contract_id	CHAR(36)	YES	FK	Reference to contracts.id
milestone_id	CHAR(36)	YES	FK	Reference to milestones.id
type	ENUM	NO	IDX	escrow_fund/milestone_release/withdrawal/refund
amount	DECIMAL(10,2)	NO		Transaction amount
fee	DECIMAL(10,2)	NO		Platform fee
net_amount	DECIMAL(10,2)	NO		Amount after fees
status	ENUM	NO	IDX	pending/completed/failed
stripe_payment_intent_id	VARCHAR(100)	YES		Stripe payment intent ID
stripe_transfer_id	VARCHAR(100)	YES		Stripe transfer ID
description	VARCHAR(255)	YES		Transaction description
created_at	DATETIME	NO	IDX	Record creation
updated_at	DATETIME	NO		Last update

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3. Indexes and Constraints

3.1 Primary Keys

All tables use UUID primary keys (CHAR(36)) for distributed system compatibility and security.

3.2 Foreign Key Constraints

Table	Column	References	On Delete
freelancer_profiles	user_id	users.id	CASCADE
client_profiles	user_id	users.id	CASCADE
projects	client_id	users.id	CASCADE
projects	category_id	categories.id	SET NULL
proposals	project_id	projects.id	CASCADE
proposals	freelancer_id	users.id	CASCADE
contracts	project_id	projects.id	CASCADE
contracts	proposal_id	proposals.id	CASCADE
milestones	contract_id	contracts.id	CASCADE
transactions	user_id	users.id	CASCADE
messages	conversation_id	conversations.id	CASCADE
messages	sender_id	users.id	CASCADE

3.3 Performance Indexes

Table	Index Name	Columns
users	idx_users_email	email
users	idx_users_role	role
projects	idx_projects_status	status
projects	idx_projects_category	category_id
projects	idx_projects_created	created_at DESC
projects	ft_projects_search	title, description (FULLTEXT)
proposals	idx_proposals_project	project_id
proposals	idx_proposals_freelancer	freelancer_id
contracts	idx_contracts_client	client_id
contracts	idx_contracts_freelancer	freelancer_id
messages	idx_messages_conversation	conversation_id, created_at
notifications	idx_notifications_user	user_id, is_read, created_at

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4. SQL Creation Scripts

Sample SQL scripts for creating the database schema:

4.1 Database Creation

```
CREATE DATABASE freelance_marketplace
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;
```

4.2 Users Table

```
CREATE TABLE users (
  id CHAR(36) PRIMARY KEY,
  email VARCHAR(254) NOT NULL UNIQUE,
  username VARCHAR(150) NOT NULL UNIQUE,
  password VARCHAR(128) NOT NULL,
  role ENUM('freelancer','client','admin') NOT NULL,
  is_active BOOLEAN DEFAULT TRUE,
  is_verified BOOLEAN DEFAULT FALSE,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  INDEX idx_users_email (email),
  INDEX idx_users_role (role)
) ENGINE=InnoDB;
```

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